

Replacing Phase 1 with Auction-Dynamics MBO

Approach B, measured against the validated deliverables — and a sixth occasion on which an instrument that could not see an effect reported its absence

August 2026

Abstract

Phase 1 (Γ -convergence relaxation by projected gradient descent) costs 22 h–79 h for a completed $N=300$ solution, and its winner-take-all readout is not a valid partition at that scale, so it is repaired afterwards. This report measures **approach B** — volume-constrained multiphase threshold dynamics [15, 7], with the volume constraint imposed by balanced assignment in the manner of auction dynamics [11] — as an outright *replacement* for Phase 1. Four results. (1) At the pinned Phase 2 campaign, B reaches perimeter **184.4118** and **184.1615** at $N=100$ (two seeds) against the control’s 185.2546, and **319.9428** at $N=300$ against 323.3192 — from 214–248 s of Phase 1 against 48,132 s and 79,069 s, a 211–319 $\times$ reduction in the replaced stage. (2) B produces **0 fragmented cells** at both N on its raw labels, where raw PGD at $N=300$ produces 2 and requires the balanced readout; connectivity is the only one of the three validity gates that has content for B, and it is the one B could have failed. (3) The win is attributable to the MBO dynamics and not to the balanced geodesic initialisation B starts from — which is approach C’s first step — by +2.761% against a pre-registered 0.3% threshold. (4) Of thirteen predictions committed before any run, nine held, three missed and one was initially mis-scored. The methodological content is again larger than the technical: a **three-part over-merge instrument was built and detected nothing**, and a per-level claim entered the record having never been measured — the sixth instance of this project’s signature artefact.

Provenance

Status	measured
Date	2026-08-19
Surface	torus $T(R=1.0, r=0.6)$ throughout
Branch	feat/phase1-mbo-auction-dynamics
Code	src/partition/mbo_auction.py (the arm); scripts/run_mbo_arm.py (driver); scripts/score_mbo_arm.py (mechanical scorer); testing/test_mbo_auction.py (G1–G8, NC1–NC5b)
Scored runs	results/arm_mbo_20260819_173411_npart100_V114144_seed84172851, results/arm_mbo_20260819_185131_npart100_V114144_seed61803399, results/arm_mbo_20260819_192659_npart300_V47488_seed61803399, results/arm_init_20260819_181203_npart100_V114144_seed84172851
Anchors	$N=100$: run_20260709_081548, Phase 2 best 185.2546144718457 (iterate 20 of 20), <i>raw</i> PGD, Phase 1 48132.12s. $N=300$: run_20260806_123326, Phase 2 best 323.319247087254 (iterate 18 of 19), PGD <i>plus</i> balanced readout, Phase 1 79068.56s. Both from timing_profile.yaml total_wall_s, never run_time.seconds.
Pre-registration	commit 9ffdefb, 2026-08-19 16:53, <i>before</i> the first scored run at 17:34:11. Results commits b414b67, c8aea83, 5bdc947, 54b937c; corrections d4dc66d, d4b75d7.
Figures	make_figures.py from the committed data.yaml, itself re-derivable by extract_data.py.
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchor check	the harness reproduces the $N=100$ control’s own perimeter <i>bitwise</i> (report 07, 0d); no Phase 2 code has changed since.

1 What B is, and what is not claimed for it

Threshold dynamics originates with Merriman, Bence and Osher [15]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and interfaces move by approximate mean curvature. Esedoğlu and Otto [7] placed the multiphase case on a variational footing with a Lyapunov functional that Γ -converges to a weighted perimeter — the same limit this project’s own energy targets [4, 17, 16]. Jacobs, Kim and Léger [11] impose *exact volume constraints* by replacing the pointwise threshold with a balanced assignment solved by Bertsekas’ auction algorithm [2]. **None of this is invented here.**

Per mesh level, the arm iterates on a *hard* labelling $\omega : V \rightarrow \{0, \dots, N-1\}$:

1. **diffuse**: solve $(M + \tau K) y_k = M \chi_k$ for each cell, with one prefactorised LU per level and N back-substitutions;
2. **balanced threshold**: $\omega(i) = \arg \max_k [y_{ik} + \psi_k]$, with ψ from the shared semi-discrete optimal-transport dual [1, 14, 12] that approach A already ships.

There is never a continuous field whose winner-take-all readout can diverge from what was constrained. B therefore does not *close* the continuous–discrete gap that motivated approaches A–E; the category is absent.

The regime, and the novelty claim we do *not* make. Every published implementation of this machinery we are aware of convolves on a *uniform Cartesian grid via FFT*, on a flat domain, at small N , and for Pólya’s *outer* problem — which convex body maximises the length of the shortest partition. That is true of Hu, Liu and Wang [10], whose auction-plus-thresholding machinery is closest to ours but whose domains are $[0, 1]^2$ and 3D boxes on $128^2/256^2$ grids, and

of Bogosel and Oudet [3], whose capacity-constrained Voronoi construction with > 100 cells is planar. Ours is the *inner* problem: partition a fixed closed genus-1 surface into N equal-area minimal-perimeter cells on an unstructured triangulated mesh, at N in the hundreds. **We phrase this as “we are not aware of”, never as “first”**: two web searches and two papers is not a literature review, and surface-remeshing and computer-graphics venues may cover adjacent ground under different vocabulary. Stated as a population claim, the corpus is: two full-text readings plus this project’s standing bibliography.

The difference is load-bearing rather than cosmetic. The FEM operator $(M + \tau K)^{-1}M$ is a different object from an FFT convolution, with different failure modes; mesh pinning [18] is ours to manage, and no grid-derived τ heuristic is imported.

2 Result 1 — perimeter

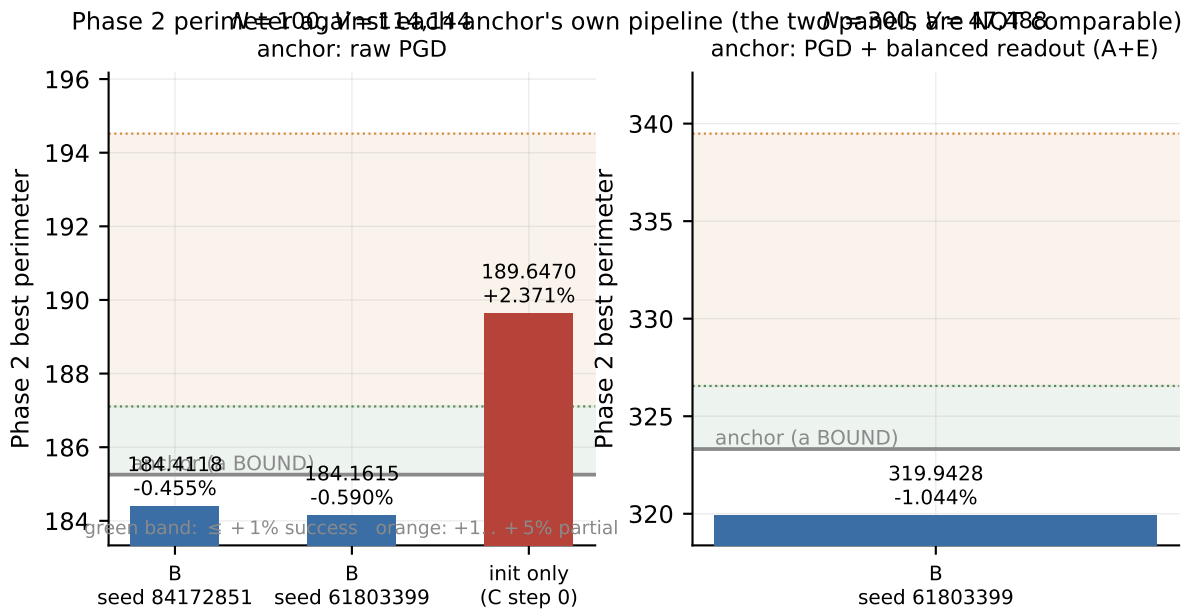


Figure 1: Phase 2 best perimeter against each anchor’s own pipeline. **The two panels are not comparable and must never be pooled**: the $N=100$ anchor is raw PGD, the $N=300$ anchor is PGD *plus* the balanced readout, because raw PGD is not a valid partition at $N=300$.

run	Phase 1	speedup	area worst	frag.	Phase 2 best	vs anchor
$N=100$ s84172851	228 s	210.9×	0.1504%	0	184.4117703525215 (18/20)	−0.455%
$N=100$ s61803399	214 s	225.2×	0.1493%	0	184.1614647856558 (19/20)	−0.590%
$N=300$ s61803399	248 s	318.8×	1.4172%	0	319.9427628182375 (16/19)	−1.044%
init only (C step 0)	67 s	—	0.1355%	0	189.6469917457051 (20/20)	+2.37%

Seed spread at $N=100$ is **0.136%**, which is $3.3\times$ smaller than the margin by which the *worse* of the two seeds beats the control. This is what makes the result scoreable at all: the pre-registration committed that a spread above 1% would mean the primary falsifier could not be evaluated at $\pm 1\%$ and that a single-seed number would be conflating method difference with seed noise.

Six constraints on how these numbers may be stated.

1. **Never pool the two anchors.** They are different baselines. “B beats the best available PGD pipeline at each N , at equal Phase 2 budget” is valid; “B beats PGD by 0.455–1.044%” as a trend in N is not.
2. **The speedup is for the replaced stage only.** End-to-end, including the shared 30–40 min Phase 2, it is $\approx 23\times$ ($N=100$) and $\approx 31\times$ ($N=300$). Against the structure-trigger PGD variant (34,967 s), the incumbent-best figure is $\approx 153\times$.
3. **Neither trajectory is a converged optimum.** Both are at the documented migration-cycling plateau. An earlier draft said “B converged where the control had not”; the sixth review resumed the control’s own Phase 2 for **20 further iterations and found no improvement** (best over all 40 remains 185.2546 at iterate 20; the extension drifts up to ≈ 185.30). B’s plateau is simply 0.455% lower. *That extension is the review’s measurement and has not been independently reproduced by us.*
4. **PGD’s own seed variance is unmeasured.** Every $N=100$ PGD run across both working directories is seed 84172851, and every $N=300$ $\lambda=11.5$ run is seed 61803399. Each anchor is a single trajectory. “B beats PGD by more than PGD’s own seed lottery” is *not* established.
5. **The $N=300$ margin cannot be decomposed, even in principle.** The readout costs +1.882% in label-boundary length (189.7310 raw $\rightarrow$ 190.7114 shifted $\rightarrow$ 193.3024 repaired, the last being what the anchor’s Phase 2 consumed). B’s raw boundary is 187.2574: shorter than the repaired labels by 3.127% *and* shorter than raw PGD’s own invalid labels by 1.304% — the strong form, since imbalance can only make a boundary artificially short. So the margin is not merely “not paying the repair bill”. But how much is un-erased repair residue cannot be separated, because **no valid repair-free PGD partition exists at $N=300$.**
6. $N=300$ rests on one Phase-2-scored seed.

3 Result 2 — validity, and which gate actually has content

“All three gates pass” is true and misleading

An earlier write-up of this result — ours — said B’s raw labels pass all three validity gates. Phase 0’s own findings say why that is the wrong emphasis:

- **Dormant is vacuous** for any balanced-assignment arm. One-hot labels give peak density 1.0 by construction; the gate cannot fail. The harness already emitted `vacuous_for_arms: True`.
- **Area is near-expected.** Equal area is precisely what B’s assignment step *optimises*. Passing means the solver converged; Phase 0 defined it as the *solver-failure detector*, not a validity test.
- **Connectivity is the only gate genuinely at risk.** Nothing in B enforces it, and the source proposal states it is “not per-step guaranteed”.

The defensible claim is narrower and still strong: **B produces 0 fragmented cells at $N=300$, where raw PGD produces 2** (with 10 imbalanced cells, worst 36.15%) — which is why PGD’s baseline required the balanced readout and B required nothing.

Re-verified through `testing/check_fragmentation.py`, which shares no code with the driver’s reporting: all four arm solutions pass, with **0 cells having more than one raw component including sub-threshold speckle**.

Robustness. Three further $N=300$ MBO ladders at seeds 27182818, 13131313 and 84172851 give **0 fragmented** in every case, with worst area 1.2531/1.4467/1.3887% and label boundaries 186.08/186.55/186.90 — all *shorter* than the scored seed’s 187.2574. Ladder only, no Phase 2; *review-produced, not independently reproduced by us.*

4 Result 3 — attribution: is this B, or is it C’s first step?

B is initialised from one balanced C-scores assignment — farthest-point seeds, multi-source Dijkstra, scores $-d^2$, balanced assignment — which is literally approach C’s step 0 [13, 1, 6]. If MBO barely improved on it, the number in §2 would be a result about C reported under B’s name, and *nothing* in the gate suite would have detected that.

	Phase 1	label boundary	Phase 2 best	censored
init only (C step 0)	67 s	116.1989	189.6469917457051 (20/20)	yes
B (MBO)	228 s	107.6014	184.4117703525215 (18/20)	no
control (PGD)	48,132 s	107.3410	185.2546144718457 (20/20)	plateaued

$$\text{P13} = \frac{189.646992 - 184.411770}{189.646992} = +\mathbf{2.761\%} \quad \text{against a pre-registered threshold of 0.3\%}.$$

Two disclosures. The measured 2.761% is an **upper bound** on MBO’s contribution, because the init arm is censored: its converged value is ≤ 189.6470 , and a lower true value shrinks the gap. For attribution to fail the init would have to reach ≈ 185.0 — another $\approx 2.5\%$ on a trajectory gaining ≈ 0.038 per iterate and decelerating. And the init-only control was run on the *finest* mesh, i.e. the *strongest* member of the init family, which biases the test *against* B.

An unflattering free datum on C. C’s step 0 plus Phase 2 lands at +2.37% above the PGD control — a PARTIAL verdict under this project’s own bands — for 67 s of Phase 1. Real C iterates Lloyd and would improve on this, so it is a *lower bound* on C; but C’s starting point is behind both B and PGD, which is directly relevant to whether C is worth building. Note also that C minimises a *quantisation* energy, not perimeter; the two optima coincide only asymptotically and only in the plane [9, 8], and this torus has Gaussian curvature of both signs.

5 The mesh ladder, and τ

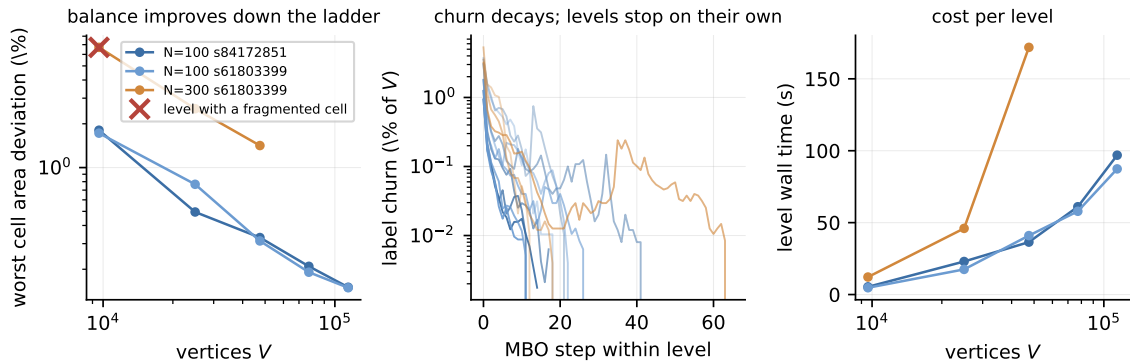


Figure 2: Per-level behaviour. Left: worst cell area deviation improves monotonically down the ladder; the cross marks the one level carrying a fragmented cell. Middle: label churn decays within every level, so levels terminate on their own rather than at the iteration cap. Right: cost per level.

$\tau = \min((c h_{\text{mean}})^2, (\rho R_{\text{cell}})^2)$ with $c = 4$ and $\rho = 1.0$, pinned before any run. $c = 4$ rather than $c = 2$: the latter is the freeze regime [18]. The cap is the over-merge side of a two-sided window, and it binds on exactly one level in use, $N=300$ level 0.

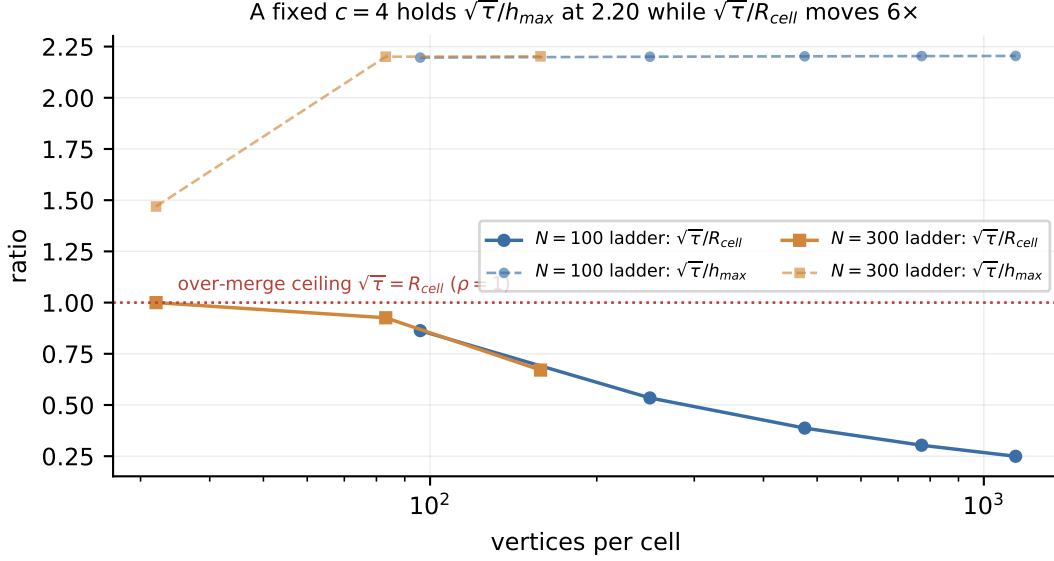


Figure 3: A fixed c does *not* hold the window. $\sqrt{\tau}/h_{\text{max}}$ stays at 2.20 across every level of both ladders while $\sqrt{\tau}/R_{\text{cell}}$ moves $6\times$. Reporting on h_{mean} alone flatters the coarse band: this mesh is $1.81\times$ anisotropic.

6 The headline null: an instrument that detected nothing

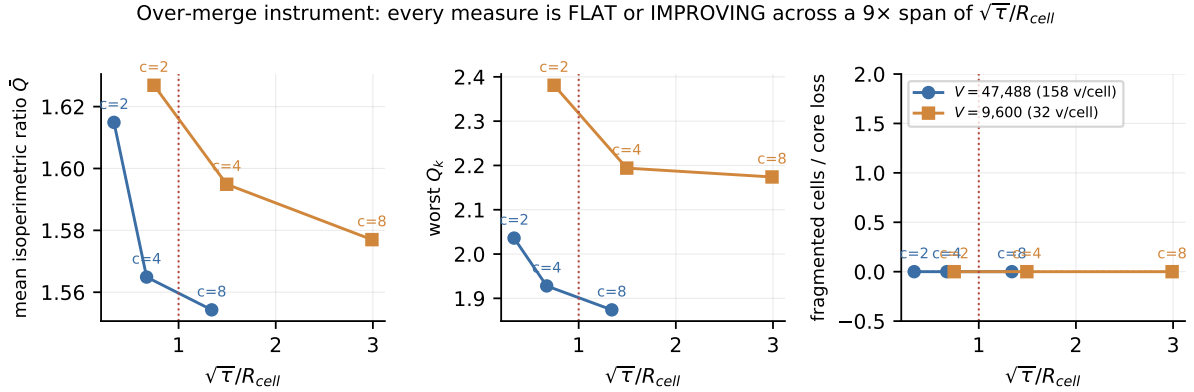


Figure 4: Fragmentation, per-cell isoperimetric ratio $Q_k = P_k^2/(4\pi A_k)$, and core loss, swept over $c \in \{2, 4, 8\}$ at two resolutions. Every measure is flat or *improving*; none registers an over-merge penalty anywhere, including at $\sqrt{\tau}/R_{\text{cell}} = 2.99$, three times the pinned cap.

Assignment quality *cannot* see over-merging — the balanced assignment hits its area target by construction, so a smeared, non-local cell scores as well as a compact one. (An earlier draft of the plan cited a healthy assignment-quality fixture as evidence against over-merging; that was a measurement of a different quantity than the claim it supported, and it was withdrawn.) So three geometry instruments were built. Across a $9\times$ span of $\sqrt{\tau}/R_{\text{cell}}$ and two levels differing

5× in resolution, **not one of them registered anything**: core loss and fragmentation are 0 in all six configurations, and compactness moves monotonically the *other* way.

The instruments are demonstrably *responsive* — $\bar{Q}$ moves 3.8%, Q_{worst} 8%, boundary length 1.9% — they simply move toward improvement. What is established is therefore that **no damage is detectable**, *not* that over-merging cannot occur.

Standing limit

The ρ question is **open**, and no future claim about c or ρ may lean on this report’s evidence in either direction. The cap was retained because it is pre-registered and because it biases *against* B — it lowers c to 2.68 at $N=300$ level 0, and $c=2$ measures worse than $c=4$. That is a discipline argument, not an empirical one. **Named follow-up**: $c=8$ showed no penalty on any instrument *and* a 0.33% shorter starting boundary — a third of the entire +1% success bar — so the cap may be costing perimeter for no measured benefit. That is a separate arm, scored through the same harness.

7 Why there is no exactly-monotone energy, and what is gated instead

Esedoğlu–Otto monotonicity is a minorise–maximise argument requiring (i) the same symmetric positive semi-definite form in the energy and in the assignment objective, and (ii) an *exact* assignment. **Both fail here, for independent reasons.** The theorem’s object is $MA_\tau = M(M + \tau K)^{-1}M$, symmetric PSD; our step maximises $\langle \chi', Dy \rangle$ with $D = \text{diag}(v)$ *lumped*, and DA_τ is not symmetric. And Jacobs–Kim–Léger obtain a theorem because Bertsekas’ auction is exact, whereas `solve_dual_offsets` is subgradient ascent returning its best iterate. **This is the price of a substitution made deliberately in Phase 0** — where Sinkhorn [5] and auction were both measured and both lost to the incumbent on assignment quality — paid here rather than there.

What survives needs neither concavity nor exactness. Since $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$, multiplying the pointwise optimality by $v_i > 0$ and summing gives

$$\langle \chi', Dy \rangle - \langle \chi, Dy \rangle \geq \sum_k \psi_k (T_k - T'_k)$$

whose right-hand side is exactly the slack that inexactness costs; it vanishes at perfect balance. Measured slack runs 1.7×10^{-4} to 5.5×10^{-3} — five to seven orders of magnitude above the 10^{-9} tolerance an earlier draft proposed for energy monotonicity, so that tolerance was never defensible. **Real E_τ increases do occur** (4 of 96 active steps at $N=300$), and gate G4 therefore gates the inequality, whose measured violation is $\leq 6.13 \times 10^{-8}$, while E_τ is reported as a descriptive trend only.

8 Predictions, and what they caught

	PREDICTED (committed 9ffdefb)	MEASURED	
P1	188.0 (+1.5%), PARTIAL	184.4118 / 184.1615, SUCCESS	MISS
P2	spread < 0.5%	0.136%	HELD
P3	<= 0.28%, point 0.15%	0.1504 / 0.1493%	HELD
P4	<= 3, point 0	0	HELD
P5	>= 30x, point 53x	210.9 / 225.2x	HELD (point missed)
P6	331.4, 80% CI [323.3, 341.0]	319.9428	MISS
P7	<= 2.02%, <= 6 frag, >= 88x	1.4172%, 0, 318.8x	HELD
P8	<= 2 of 5 flagged	0, clauses agree	HELD
P8'	dE flags L2,L3,L4 (falsifying P8)	dE flat 14-22, flags none	MISS
P9	frag not death; worst at L0	1 frag at L0 (295), 0 elsewhere	HIT
P10	viol <= 1e-12; excursions <= ceiling	-6.13e-08; 4/96 within ceiling	HELD
P12	no abort	20/20, 19/19	HELD
P13	>= 0.3%	+2.761%	HELD

Figure 5: Thirteen predictions committed in 9ffdefb at 16:53, before the first scored run at 17:34:11.

P1 and P6 missed, and P1’s *reasoning* is refuted. The rationale was that B sees perimeter only through a coarse- τ surrogate while PGD saw the actual energy for 13.4 h. B also *enters* Phase 2 from a 0.243% *longer* boundary than the control and still finishes ahead — a genuine crossover, so the advantage is not initial geometry. That it is combinatorial is a hypothesis consistent with these data, not established by them. P6 missed worse: its 80% interval’s lower bound was the anchor itself, so essentially no probability was assigned to B beating PGD + A + E at $N=300$.

P8’ missed and invalidated its own model. It predicted, from a log-linear interpolation of $\Delta E/\text{tail}$ in the freeze ratio c/c_{lo} , that levels 2–4 would be flagged pinned. Measured, $\Delta E/\text{tail}$ is *flat* across the ladder — 15.76, 21.62, 15.11, 14.22, 16.49 against a threshold of 79.25 — and level 4, sitting exactly at $c/c_{lo} = 0.999$, scores 16.49 where the model demanded ≈ 260 . The freeze ratio does not predict pinning here.

9 The sixth artefact — and it is ours

Report 07 catalogued five occasions on which a measurement artefact was read as a result about the method. This is the sixth, and it was produced in this work.

The plan asserted “**0 fragmented at every level**” and, on that basis, scored P9 — which predicted over-merge would appear as fragmentation rather than cell death, worst at level 0 — as UNTESTABLE, “the predicted phenomenon never occurred”. **Until the sixth review, LevelReport carried no connectivity field at all.** Per-level fragmentation had never been computed. The claim was an inference presented as a result, and it was false.

With the instrument added and the $N=300$ ladder re-run to a bit-exact reproduction (worst areas 6.7063/2.5394/1.4172%, steps 16/21/66, boundary 187.2574):

level 0: 1 fragmented cell (cell 295). Levels 1 and 2: 0. No cell death.

The phenomenon occurred, at exactly the level P9 named, in exactly the form P9 predicted. **P9 is a hit, not untestable** — “untestable” had absorbed a correct prediction on the strength of a measurement never taken. The stray is healed by level 1’s MBO run, which is direct evidence for the mechanism the taxonomy predicted: stray islands are perimeter-inefficient and shrink under curvature flow, the opposite dynamic to the discrete-area trim that manufactured them.

The fix is instrumentation — `run_gates` per level, seconds of cost, now committed — not a resolution to be careful.

10 Threats to validity that no instrument here addresses

1. **PGD’s seed variance** (§2, constraint 4). Both anchors are single trajectories; the corpus was enumerated across *both* working directories, since `results/` is gitignored and each worktree has its own.
2. **Over-merging** (§6). Not excluded, merely undetected.
3. **Phase 2 budget as a confound**. B may start from geometry that lets Phase 2 converge *within* a budget PGD cannot use as well. The 40-iteration control extension argues against this, but is review-produced.
4. **Same-machine identity** of the July control and the August arms is consistent with the recorded hardware but is not stamped in run metadata.
5. $N=1000$ is untouched. Nothing here establishes that B’s behaviour extrapolates.

11 Conclusions

1. B replaces Phase 1 at $211\text{--}319\times$ less wall time in the replaced stage and reaches a *shorter* Phase 2 perimeter than the best available PGD pipeline at both N , at equal Phase 2 budget.
2. Its raw labels are connected at both N — the one validity gate B could have failed — so the readout and repair stages are unnecessary for B.
3. The advantage is attributable to the MBO dynamics, not to the balanced geodesic initialisation, by $+2.761\%$ against a 0.3% threshold.
4. Three pre-registered predictions missed, all in B’s favour, and one was initially mis-scored in B’s *disfavour* by an instrument that did not exist.

References we could not verify

Author given names for [10] are **unverified**; the entry records “Hu, Wei and Liu, Chong and Wang, Dong” from a secondary description, not from the paper’s title page. Volume and page numbers throughout `docs/math/shared/references.bib` carry an upstream note that they should be checked against published sources before external distribution, and that note still stands. Neither [10] nor [3] was read by the report’s author; both are summarised from a literature review supplied to this work, and the *characterisation* of their scope (outer problem, flat Cartesian domains, small N) rests on that review.

Related documents

- Plan, pre-registration and full correction history: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The instrument this report is scored through: `docs/experiments/07-phase0-shared-harness/`
- The gap B makes vacuous: `docs/reference/winner_take_all_partition_gap.md`

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